THE MASTERMIND COACHING CLASSES

Revision Notes

Class 10 · Mathematics

Chapters 1 to 5 · NCERT Mathematics — Class 10 · CBSE / RBSE Syllabus 2026-27

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
1	Real Numbers	Fundamental Theorem of Arithmetic, HCF and LCM, irrational numbers, decimal expansions
2	Polynomials	Zeroes of a polynomial, graphs, relation between zeroes and coefficients of a quadratic
3	Pair of Linear Equations in Two Variables	Graphical method, consistency, substitution and elimination, word problems
4	Quadratic Equations	Factorisation, completing the square, quadratic formula, discriminant, applications
5	Arithmetic Progressions	nth term and sum of n terms of an AP, applications

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CHAPTER 01 · MATHS

Real Numbers

Every number you have met so far, whole numbers, fractions, decimals and roots, lives on the number line as a real number. This chapter looks deeper at how composite numbers are built from primes, how that idea gives the HCF and LCM, how we prove that numbers like the square root of 2 are irrational, and how to tell from a fraction whether its decimal stops or repeats.

Big Questions

- Can every composite number be written as a product of primes in only one way?
- How do prime factors give the HCF and the LCM?
- How do we prove a number is irrational, and when does a decimal terminate?

1.1 Fundamental Theorem of Arithmetic

DEFINITION

Fundamental Theorem of Arithmetic: every composite number can be expressed (**factorised**) as a product of primes, and this factorisation is **unique**, except for the order in which the prime factors occur.

For example $3825 = 3^2 \times 5^2 \times 17$, and no other set of primes multiplies to 3825. The theorem is used to decide whether a number can end in a given digit: for 4^n to end in 0 it would need the prime 5 as a factor, but $4^n = 2^{2n}$ has only the prime 2. So **4^n never ends in 0**; similarly $(2 \times 3)^n = 6^n$ has no factor 5.

1.2 HCF and LCM by Prime Factorisation

Step 1 — Factorise each number
Write every number as a product of prime powers.

Step 2 — HCF
Take each **common** prime with the **smallest** power.

Step 3 — LCM
Take **every** prime that appears with the **greatest** power.

Numbers	Prime factors	HCF	LCM
6, 72	2×3; $2^3 \times 3^2$	$2 \times 3 = 6$	$2^3 \times 3^2 = 72$
96, 404	$2^5 \times 3$; $2^2 \times 101$	$2^2 = 4$	$2^5 \times 3 \times 101 = 9696$
6, 72, 120	2×3; $2^3 \times 3^2$; $2^3 \times 3 \times 5$	$2 \times 3 = 6$	$2^3 \times 3^2 \times 5 = 360$

FORMULA

For two positive integers a and b: **HCF(a, b) x LCM(a, b) = a x b.** This does **not** hold for three numbers. HCF always divides the LCM.

Check: $96 \times 404 = 38{,}784$ and $4 \times 9696 = 38{,}784$. Given HCF = 9 and LCM = 360 with one number 45, the other is $(9 \times 360)/45 = \mathbf{72}$.

Word problem. Three bells ring every 6, 8 and 12 minutes. They next ring together after the **LCM** = 24 minutes. Square tiles of the largest size covering a 120 cm x 84 cm floor without cutting have side equal to the **HCF** = 12 cm.

1.3 Irrational Numbers

DEFINITION

A number is **rational** if it can be written as p/q with integers p and q (q not equal to 0), and **irrational** if it cannot. Irrational numbers have decimals that are non-terminating and non-repeating.

Theorem: if a prime p divides a^2, then p divides a. It is used to prove that the square roots of primes are irrational.

Proof that root 2 is irrational

Assume root 2 = a/b with a and b co-prime and b not 0.

Square

$2b^2 = a^2$, so 2 divides a^2, hence 2 divides a. Write a = 2c.

Contradiction

Then $b^2 = 2c^2$, so 2 divides b as well. Both are divisible by 2, which contradicts that a and b are co-prime. Hence **root 2 is irrational**.

- The same method proves root 3 and root 5 are irrational.
- **Sum, difference, product or quotient of a non-zero rational and an irrational is irrational:** 3 + 2 root 5, 5 − root 3 and 7 root 2 are irrational. (Assume 3 + 2 root 5 = r, rational; then root 5 = (r − 3)/2 would be rational, a contradiction.)
- The sum or product of two irrationals can be rational: root 2 x root 2 = 2 and (3 + root 2) + (3 − root 2) = 6.

1.4 Decimal Expansions of Rational Numbers

FORMULA

Let x = p/q in lowest terms. If **$q = 2^m 5^n$** (only 2 and 5 as prime factors) the decimal **terminates**. If q has any other prime factor, the decimal is **non-terminating and repeating**.

Rational number	Denominator	Decimal
13/3125	5^5	Terminates (0.00416)
17/8	2^3	Terminates (2.125)
64/455	5 x 7 x 13	Non-terminating repeating
7/6	2 x 3	Non-terminating repeating (1.1666...)
29/343	7^3	Non-terminating repeating

IMPORTANT

Reduce the fraction first: 6/15 = 2/5 has denominator 5, so it terminates, while 35/50 = 7/10 also terminates. After division, a terminating decimal has as many decimal places as the larger power of 2 or 5 in q.

1.5 Worked Examples

Show that 5 − root 3 is irrational. If 5 − root 3 = r (rational) then root 3 = 5 − r would be rational, but root 3 is irrational. Contradiction.

Can 12^n end in the digit 0? $12^n = 2^{2n} \times 3^n$ has no prime factor 5, so it cannot.

Is 7 x 11 x 13 + 13 composite? It equals 13(77 + 1) = 13 x 78, a product of two factors greater than 1, so it is composite.

Quick Revision: Likely Exam Questions

Likely question	Model answer
State the Fundamental Theorem of Arithmetic.	Every composite number is a product of primes, and the factorisation is unique apart from order.
Find the HCF and LCM of 12 and 18 by prime factorisation.	$12 = 2^2 \times 3$, $18 = 2 \times 3^2$: HCF = 6, LCM = 36.
If HCF(a,b) = 6 and a x b = 1800, find LCM.	1800/6 = 300.
Is 0.1010010001... rational?	No, it is non-terminating and non-repeating.
Without dividing, does $23/(2^3 \times 5^2)$ terminate?	Yes; the denominator has only 2 and 5.
Prove root 5 is irrational.	Assume a/b in lowest terms, then 5 divides a^2, so 5 divides a, then 5 divides b: contradiction.
Is (2 + root 3)(2 − root 3) rational?	Yes, it equals 4 − 3 = 1.
Write the smallest prime and the smallest composite number.	2 and 4.

★ KEY TAKEAWAYS

- Every composite number has a **unique prime factorisation**.
- **HCF** = smallest powers of common primes; **LCM** = greatest powers of all primes; **HCF x LCM = a x b** for two numbers.
- Root of a prime, and (rational ± irrational), are **irrational**; prove by contradiction.
- p/q terminates if and only if the denominator in lowest terms is 2^m5^n.

CHAPTER 02 · MATHS

Polynomials

Polynomials describe the path of a thrown ball, the shape of a bridge cable and the cost of a product depending on its size. A zero of a polynomial is a value of the variable that makes it vanish, and zeroes can be read off graphs. This chapter links zeroes to the coefficients of quadratic polynomials so that we can build a polynomial from its zeroes.

Big Questions

- What is a zero of a polynomial and how is it seen on a graph?
- How many zeroes can a linear, quadratic or cubic polynomial have?
- How are the zeroes of a quadratic related to its coefficients?

2.1 Polynomials and Their Degree

Degree	Name	General form	Example
1	Linear	$ax + b$	$3x - 5$
2	Quadratic	$ax^2 + bx + c$	$x^2 - 3x + 2$
3	Cubic	$ax^3 + bx^2 + cx + d$	$x^3 - 4x$

DEFINITION

A **zero** of a polynomial p(x) is a real number k such that **p(k) = 0**. The **degree** is the highest power of x. A polynomial of degree n has **at most n zeroes**.

Example. For $p(x) = x^2 - 3x - 4$: $p(4) = 16 - 12 - 4 = 0$ and $p(-1) = 1 + 3 - 4 = 0$, so 4 and −1 are the zeroes.

2.2 Geometrical Meaning of Zeroes

The real zeroes of p(x) are the **x-coordinates of the points where the graph of y = p(x) meets the x-axis**.

Polynomial	Shape of graph	Possible number of zeroes
Linear ax + b	Straight line (not parallel to x-axis)	Exactly one: −b/a
Quadratic $ax^2 + bx + c$	Parabola; opens **up if a > 0**, **down if a < 0**	0 (no intersection), 1 (touches), or 2
Cubic	Curve with up to two turns	1, 2 or at most 3

- If the parabola cuts the x-axis at two points there are two distinct zeroes; if it just touches there is one (repeated) zero; if it stays above or below the axis there are none.
- **$x^2 + 1$** has no real zero as its graph is above the x-axis; **$x^3 - x$** has three zeroes (−1, 0, 1).

2.3 Relation between Zeroes and Coefficients

FORMULA

If α and β are the zeroes of $\mathbf{ax^2 + bx + c}$: $\mathbf{\alpha + \beta = -b/a}$ and $\mathbf{\alpha\beta = c/a}$. A quadratic with zeroes α and β is $\mathbf{k[x^2 - (\alpha + \beta)x + \alpha\beta]}$.

Polynomial	Zeroes	Sum = −b/a	Product = c/a
$\mathbf{x^2 - 2x - 8}$	4, −2	2 = −(−2)/1	−8
$\mathbf{4s^2 - 4s + 1}$	1/2, 1/2	1 = 4/4	1/4
$\mathbf{6x^2 - 3 - 7x}$	3/2, −1/3	7/6	−1/2
$\mathbf{3x^2 - x - 4}$	4/3, −1	1/3	−4/3

▸ Useful identities in α and β

- $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
- $1/\alpha + 1/\beta = (\alpha + \beta)/\alpha\beta$
- $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

2.4 Worked Examples

Find the zeroes of $x^2 - 5x + 6$ and verify. Factorise: (x − 2)(x − 3), so zeroes are **2 and 3**. Sum 5 = −(−5)/1 and product 6 = 6/1. Verified.

Form a quadratic with sum of zeroes −3 and product 2. $x^2 - (-3)x + 2 = \mathbf{x^2 + 3x + 2}$; the zeroes are −1 and −2.

If α and β are the zeroes of $x^2 - 4x + 1$, find $\alpha^2 + \beta^2$ and $1/\alpha + 1/\beta$. Sum = 4 and product = 1; $\alpha^2 + \beta^2 = 16 - 2 = \mathbf{14}$; $1/\alpha + 1/\beta = 4/1 = \mathbf{4}$.

If 2 is a zero of $x^2 - 3x + k$, then 4 − 6 + k = 0, so **k = 2**. The other zero is 3 − 2 = 1 (and product 2 matches).

Zeroes of $3x^2 - 5x - 2$: sum 5/3, product −2/3, so $\alpha + \beta + \alpha\beta = 5/3 - 2/3 = \mathbf{1}$.

From the graph. A parabola cutting the x-axis at −3 and 1 means the quadratic has two zeroes, sum −2 and product −3, so it is $k(x^2 + 2x - 3)$.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is the maximum number of zeroes of a cubic?	Three.
Find the zero of 2x + 7.	−7/2.
If the sum and product of zeroes are 0 and −4, find the polynomial.	$x^2 - 4$.
How many zeroes if the graph does not meet the x-axis?	None (no real zero).

Likely question	Model answer
Product of zeroes of $x^2 + 7x + 10$?	10.
If α + β = 5 and αβ = 6, find the zeroes.	2 and 3.
When does the parabola open downward?	When a < 0.
Is −2 a zero of $x^2 + 3x + 2$?	Yes: 4 − 6 + 2 = 0.

★ KEY TAKEAWAYS

- A zero of p(x) satisfies **p(k) = 0**; it is where the graph **cuts the x-axis**.
- Degree n gives **at most n** zeroes; a quadratic graph is a **parabola**.
- **α + β = −b/a; αβ = c/a.**
- Polynomial from zeroes: **x^2 − (sum)x + product**.

CHAPTER 03 · MATHS

Pair of Linear Equations in Two Variables

A shopkeeper's bill for two items on two days, the speeds of a boat in still water and in a stream, or the ages of a parent and child can be turned into two equations with two unknowns. This chapter shows how a pair of linear equations can have one solution, none or infinitely many, and how to solve them by drawing graphs, by substitution and by elimination.

Big Questions

- What does the graph of a pair of linear equations tell us about its solutions?
- How do the coefficients decide whether the pair is consistent?
- How do we solve and use the pair in everyday situations?

3.1 Pair of Linear Equations

A pair is written $\mathbf{a_1x + b_1y + c_1 = 0}$ and $\mathbf{a_2x + b_2y + c_2 = 0}$. Each equation represents a straight line; a **solution** is a point lying on both lines.

3.2 Graphical Method and Types of Solution

Lines	Condition on coefficients	Number of solutions	Name
Intersecting	$a_1/a_2 \neq b_1/b_2$	Exactly one	**Consistent** (independent)
Coincident	$a_1/a_2 = b_1/b_2 = c_1/c_2$	Infinitely many	**Consistent** (dependent)
Parallel	$a_1/a_2 = b_1/b_2 \neq c_1/c_2$	None	**Inconsistent**

DEFINITION

A pair is **consistent** if it has at least one solution and **inconsistent** if it has none. For the graphical method, find two or three points of each line, plot them and read the intersection.

Example. x + y = 5 and 2x − y = 4: lines meet at (3, 2), so x = 3, y = 2. Here 1/2 ≠ 1/(−1): intersecting. The pair 2x + 3y = 9 and 4x + 6y = 18 are coincident; 2x + 3y = 9 and 4x + 6y = 12 are parallel (no solution).

3.3 Algebraic Methods

▸ Substitution method

Step 1
Express one variable in terms of the other from either equation.

Step 2
Substitute it in the other equation to get an equation in one variable and solve.

Step 3
Put the value back to find the second variable and check in both equations.

Example. 2x + 3y = 12 and x − y = 1. From the second, x = y + 1. Then 2(y + 1) + 3y = 12, so 5y = 10, **y = 2, x = 3**.

▸ Elimination method

Step 1
Multiply the equations by suitable numbers so that the coefficients of one variable become equal.

Step 2
Add or subtract to eliminate it and solve for the other.

Step 3
Back-substitute.

Example. 3x + 2y = 11 and 2x + 3y = 4. Multiply the first by 3 and the second by 2: 9x + 6y = 33 and 4x + 6y = 8. Subtract: 5x = 25, **x = 5**, then y = **−2**. Check: 15 − 4 = 11 and 10 − 6 = 4.

> **IMPORTANT**
> If elimination leaves a true statement such as 0 = 0 the pair has **infinitely many solutions**; a false statement such as 0 = 5 means **no solution**.

3.4 Finding the Value of k

Requirement	Use
Unique solution	$a_1/a_2 \neq b_1/b_2$
Infinitely many	$a_1/a_2 = b_1/b_2 = c_1/c_2$
No solution	$a_1/a_2 = b_1/b_2 \neq c_1/c_2$

Example. 2x + 3y = 7 and (k − 1)x + (k + 2)y = 3k have infinitely many solutions. Then 2/(k − 1) = 3/(k + 2), so 2k + 4 = 3k − 3 and **k = 7**; check 7/(3k) = 7/21 = 1/3, which equals 2/6. **Example:** kx + 3y = 3 and 12x + ky = 6 have no solution if k/12 = 3/k ≠ 1/2: $k^2 = 36$, k = 6 or −6, and k/12 ≠ 1/2 rules out 6, so **k = −6**.

3.5 Word Problems

Read carefully, name the unknowns x and y, form two equations and solve. **Digits:** the sum of the digits of a two-digit number is 9 and the number obtained by reversing the digits is 27 more than the original. With digits x (tens) and y: x + y = 9 and (10y + x) − (10x + y) = 27, so y − x = 3. Then x = 3 and y = 6, the number is **36**.

Ages. A father is three times as old as his son; in 12 years he will be twice as old. F = 3S and F + 12 = 2(S + 12) give 3S + 12 = 2S + 24, so son **12**, father **36**.

Speed (stream). A boat covers 24 km upstream and 28 km downstream in 6 hours, and 30 km upstream with 21 km downstream in 6.5 hours. With u = 1/(x − y) and v = 1/(x + y) we get speeds of boat 10 km/h and stream 4 km/h.

Fare. A taxi charges a fixed fee plus a rate per km; 10 km costs &rupee;105 and 15 km costs &rupee;155, so the rate is &rupee;10/km and fixed fee &rupee;5.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What type of lines show an inconsistent pair?	Parallel lines.

Likely question	Model answer
How many solutions: 3x + 4y = 2, 6x + 8y = 4?	Infinitely many (coincident lines).
Solve x + y = 14 and x − y = 4.	x = 9, y = 5.
Condition for a unique solution?	$a_1/a_2 \neq b_1/b_2$.
Find k for no solution: x + 2y = 3, 3x + ky = 5?	k = 6 (since 1/3 = 2/k ≠ 3/5).
Name two algebraic methods.	Substitution and elimination.
A line has equation 2x + 3y = 6; find where it meets the axes.	(3, 0) and (0, 2).
Graphically, what does the solution represent?	The point of intersection of the two lines.

★ KEY TAKEAWAYS

- Intersecting lines give **one solution**, coincident **infinitely many**, parallel **none**.
- Use ratios **a_1/a_2, b_1/b_2, c_1/c_2** to decide the type.
- Solve by **substitution** or **elimination**; always verify in both equations.
- Word problems: define two variables, write two equations, solve and state the answer with units.

CHAPTER 04 · MATHS

Quadratic Equations

The area of a rectangular plot with a given perimeter, the speed of a train on a journey, or the height of a thrown stone at a given time: such problems lead to equations in which the unknown appears squared. This chapter defines a quadratic equation, solves it by factorisation, by completing the square and by a formula, uses the discriminant to describe the nature of roots, and applies the equations to word problems.

Big Questions

- How do we solve $ax^2 + bx + c = 0$ in different ways?
- How does the discriminant decide the nature of roots?
- How do we convert a real situation into a quadratic equation?

4.1 Quadratic Equation

DEFINITION

A **quadratic equation** in x is one of the form $\mathbf{ax^2 + bx + c = 0}$, where a, b, c are real and $\mathbf{a \neq 0}$. Its solutions are called **roots** (zeroes of the polynomial); it has **at most two roots**.

Equations such as $(x - 2)^2 + 1 = 2x - 3$ reduce to $x^2 - 6x + 8 = 0$, which is quadratic; $x(x + 1) + 8 = (x + 2)(x - 2)$ reduces to $x + 12 = 0$, which is **not** quadratic.

4.2 Solution by Factorisation

Step 1

Write the equation as $ax^2 + bx + c = 0$.

Step 2

Split the middle term into two terms whose product is ac and sum is b.

Step 3

Factorise and equate each factor to zero.

Example. $2x^2 - 5x + 3 = 0$: split $-5x = -2x - 3x$: $2x(x - 1) - 3(x - 1) = (2x - 3)(x - 1) = 0$. Roots **3/2 and 1**. Another: $x^2 - 3x - 10 = (x - 5)(x + 2) = 0$; roots 5 and −2.

4.3 Completing the Square and the Formula

Example (completing the square). $x^2 + 6x - 7 = 0$: add 9 to both sides after moving the constant: $x^2 + 6x + 9 = 16$, so $(x + 3)^2 = 16$, $x + 3 = \pm 4$, giving **x = 1 or x = −7**.

FORMULA

Quadratic formula: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$, valid when $b^2 - 4ac \geq 0$. The expression $\mathbf{D = b^2 - 4ac}$ is the **discriminant**.

Example. $x^2 - 6x + 4 = 0$: $D = 36 - 16 = 20$, $x = (6 \pm 2\sqrt{5})/2 = \mathbf{3 \pm \sqrt{5}}$. Also $2x^2 - 7x + 3 = 0$: $D = 49 - 24 = 25$, $x = (7 \pm 5)/4 = 3$ or $1/2$.

4.4 Nature of Roots

Discriminant	Nature of roots	Graph (parabola)
D > 0	Two distinct real roots	Cuts the x-axis at two points
D = 0	Two equal real roots, x = −b/2a	Touches the x-axis
D < 0	No real roots	Does not meet the x-axis

Example. For $3x^2 - 4\sqrt{3}\,x + 4 = 0$: D = 48 − 48 = 0, so equal roots $x = 2/\sqrt{3}$. **Find k** for which $kx(x - 2) + 6 = 0$ has equal roots: $kx^2 - 2kx + 6 = 0$, $D = 4k^2 - 24k = 0$, **k = 6** (k = 0 rejected). For real roots of $x^2 + kx + 9$: $D = k^2 - 36 \geq 0$, so $k \leq -6$ or $k \geq 6$.

4.5 Applications

Read and assume
Let the unknown quantity be x.

Form the equation
Translate the statements into a quadratic.

Solve and reject
Reject roots that are negative, fractional or impossible, as lengths, ages and counts cannot be so.

Verify
Check against the original statement.

Problem	Equation	Answer
Product of two consecutive positive integers is 306	$x(x + 1) = 306$, $x^2 + x - 306 = 0$	x = 17: numbers 17, 18
A rectangular plot has length 3 m more than twice its breadth and area 65 m^2	$b(2b + 3) = 65$, $2b^2 + 3b - 65 = 0$	b = 5 m, l = 13 m
A train covers 360 km at a uniform speed; at 5 km/h faster it takes 1 hour less	$360/x - 360/(x + 5) = 1$, $x^2 + 5x - 1800 = 0$	x = 40 km/h
Two sisters: one is 4 years older and the product of their ages is 192	$x(x + 4) = 192$, $x^2 + 4x - 192 = 0$	12 and 16

IMPORTANT
Two roots do not always make sense: a negative speed, length or age is rejected. A “no real roots” discriminant means the situation described cannot happen.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Is $x(x + 2) = x^2 + 3$ quadratic?	No; it reduces to 2x − 3 = 0, which is linear.
Discriminant of $2x^2 - 4x + 3$?	16 − 24 = −8, so no real roots.
Roots of $x^2 - 7x + 12 = 0$?	3 and 4.

Likely question	Model answer
State the condition for equal roots.	$b^2 - 4ac = 0$.
If one root of $x^2 - 5x + k = 0$ is 2, find k.	$k = 6$.
Value of k for which $2x^2 + kx + 3 = 0$ has equal roots?	$k^2 = 24$, $k = \pm 2\sqrt{6}$.
Why must $a \neq 0$?	Otherwise the equation becomes linear.
Roots of $x^2 + 2x - 15$?	3 and −5.

★ KEY TAKEAWAYS

- Standard form **$ax^2 + bx + c = 0, a \neq 0$**; at most two roots.
- Solve by **factorisation, completing the square, or the formula**.
- **$D = b^2 - 4ac$:** positive (two real), zero (equal), negative (none).
- In word problems, **reject** roots that are not meaningful and verify.

CHAPTER 05 · MATHS

Arithmetic Progressions

Savings that grow by the same amount every month, seats in rows of an auditorium that increase by a fixed number, and taxi fares that rise by a fixed amount for each kilometre follow patterns called arithmetic progressions. This chapter finds any term of such a list and the sum of its first n terms without writing them all out.

Big Questions

- What makes a list of numbers an arithmetic progression?
- How do we find the nth term directly?
- How do we add up the first n terms quickly?

5.1 What is an AP?

DEFINITION

An **arithmetic progression (AP)** is a list of numbers in which each term is obtained by adding a fixed number **d**, the **common difference**, to the previous term. The first term is **a**, and **$d = a_2 - a_1 = a_3 - a_2 = \ldots$**

List	Is it an AP?	a and d
3, 7, 11, 15, ...	Yes	a = 3, d = 4
10, 6, 2, −2, ...	Yes	a = 10, d = −4
2, 2, 2, 2, ...	Yes (constant)	a = 2, d = 0
1, 3, 6, 10, ...	No	differences 2, 3, 4 are not equal
1/2, 3/2, 5/2, ...	Yes	a = 1/2, d = 1

- If d > 0 the AP is **increasing**, if d < 0 it is **decreasing**, if d = 0 it is constant.
- An AP can be **finite** (has a last term) or **infinite**.

5.2 The nth Term

FORMULA

$a_n = a + (n - 1)d$. The **n-th term from the end** of a finite AP with last term l is **$l - (n - 1)d$**.

Example. For 3, 7, 11, ...: $a_{10} = 3 + 9 \times 4 = 39$. **Which term of 3, 8, 13, 18, ... is 78?** 78 = 3 + (n − 1)5 gives n − 1 = 15, so it is the **16th term**. **Is 301 a term of 5, 11, 17, 23, ...?** 301 = 5 + (n − 1)6 gives n − 1 = 49.33, not an integer, so **no**.

Finding the AP from conditions. If $a_3 = 16$ and $a_7 - a_5 = 12$: 2d = 12, d = 6, a + 12 = 16, a = 4. The AP is **4, 10, 16, 22, ...**. **Number of terms:** in 10, 7, 4, ..., −62: −62 = 10 + (n − 1)(−3), n − 1 = 24, so **25 terms**.

5.3 Sum of the First n Terms

FORMULA

$S_n = (n/2)[2a + (n - 1)d]$ and, if the last term l is known, **$S_n = (n/2)(a + l)$.** Also **$a_n = S_n - S_{n-1}$**.

Result	Value
Sum of first n natural numbers	n(n + 1)/2; for n = 100 it is 5050
Sum of first n odd numbers	n^2
Sum of first n even numbers	n(n + 1)
Sum of first 15 multiples of 8 (8, 16, ..., 120)	(15/2)(8 + 120) = 960

Example. First 20 terms of 7, 11, 15, ...: S_{20} = 10[14 + 19 x 4] = 10 x 90 = **900**. **How many terms of 24, 21, 18, ... give a sum of 78?** (n/2)[48 − 3(n − 1)] = 78, so $3n^2 - 51n + 156 = 0$, i.e. $n^2 - 17n + 52 = 0$, giving n = 4 or 13. **Both are valid**: the 9th term is 0 and later terms are negative, so the 5th to 13th terms add up to zero.

If $S_n = 3n^2 + n$: $a_1 = S_1 = 4$; $S_2 = 14$, so $a_2 = 10$ and d = 6; $a_n = 6n - 2$.

5.4 Applications

- **Seating:** rows with 20, 22, 24, ... seats: 12th row has 20 + 11 x 2 = 42 seats; 12 rows hold (12/2)(20 + 42) = 372.
- **Savings:** &rupee;500 in the first month, &rupee;50 more each month: after 12 months total = 6 x (1000 + 550) = &rupee;9300.
- **Fines or salary:** a salary rising by a fixed increment each year is an AP; a fine of &rupee;200 on the first day and &rupee;50 more each day is an AP with a = 200, d = 50.
- **Three terms in AP:** take a − d, a, a + d; four terms: a − 3d, a − d, a + d, a + 3d (sums simplify).

IMPORTANT

Check that the list is an AP before using the formulae. When a quadratic gives two values of n, discard any that are negative or not whole numbers.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Find the 10th term of 2, 7, 12, ...	2 + 9 x 5 = 47.
Find d if $a_7 = 20$ and a = 2.	2 + 6d = 20, d = 3.
Sum of first 10 terms of 5, 9, 13, ...?	5[10 + 36] = 230.
Is 2, 4, 8, 16 an AP?	No, differences 2, 4, 8 are unequal.
Find a_n if $S_n = n^2$.	$a_n = 2n - 1$.
The 5th and 9th terms of an AP are 13 and 25; find d.	4d = 12, d = 3.
Write the sum of first 50 natural numbers.	1275.

Likely question	Model answer
In an AP a = 6, l = 66, n = 11; find S_n.	(11/2)(72) = 396.

★ KEY TAKEAWAYS

- AP: constant **$d = a_{n+1} - a_n$**.
- **$a_n = a + (n - 1)d$.**
- **$S_n = (n/2)[2a + (n - 1)d] = (n/2)(a + l)$.**
- **$a_n = S_n - S_{n-1}$**; check both roots of n for validity.